Integration & Applications

Section 1: Indefinite Integrals & Power Rule

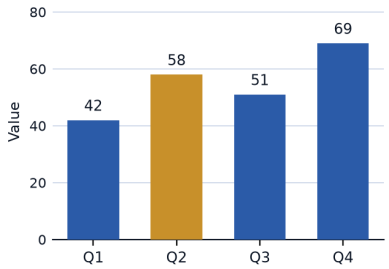
1 Evaluate the indefinite integral: $\int (6x^2 - 4x + 3)\,dx$.

a. $2x^3 - 2x^2 + 3x + C$

b. $3x^3 - 2x^2 + 3x + C$

c. $2x^3 - 4x^2 + 3x + C$

d. $6x^3 - 2x^2 + 3x + C$



2 Evaluate $\int 8x^3\,dx$.

a. $2x^4 + C$

b. $4x^4 + C$

c. $8x^4 + C$

d. $x^4 + C$

3 What is $\int \frac{1}{x^2}\,dx$?

a. $-\frac{1}{x} + C$

b. $\frac{1}{x} + C$

c. $-\frac{2}{x^3} + C$

d. $\ln(x^2) + C$

Section 2: Definite Integrals

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4 Evaluate the definite integral: $\int_0^3 2x\,dx$.

a. 9

b. 6

c. 18

d. 12

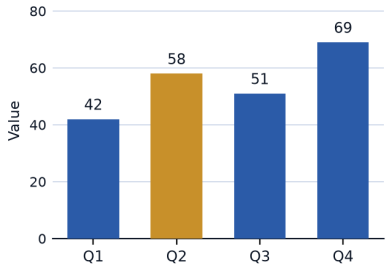
5 Evaluate $\int_1^2 (3x^2 + 1)\,dx$.

a. 8

b. 7

c. 9

d. 10



6 What is the value of $\int_{-2}^2 x^3\,dx$?

a. 0

b. 8

c. 16

d. -8

Section 3: Area Under Curves

7 What is the area of the region bounded by $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 3$?

a. 9

b. 6

c. 12

d. 27

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8 What is the area of the region under the line $y = 4$ from $x = 1$ to $x = 5$?

a. 16

b. 20

c. 12

d. 8

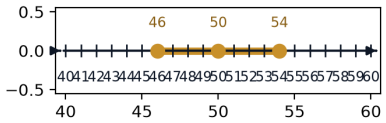
9 Find the area between the curve $y = 6x - x^2$ and the x-axis from $x = 0$ to its root $x = 6$.

a. 36

b. 18

c. 72

d. 54



Section 4: Fundamental Theorem of Calculus

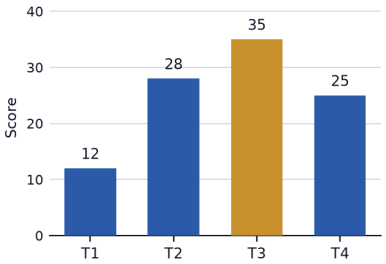
10 If $F(x) = \int_1^x (3t^2 - 2)\,dt$, what is $F'(x)$?

a. $3x^2 - 2$

b. $x^3 - 2x$

c. $6x$

d. $3x^2$



11 If $\int_1^4 f(x)\,dx = 7$ and $\int_2^4 f(x)\,dx = 3$, what is $\int_1^2 f(x)\,dx$?

a. 4

b. 10

c. 2

d. 5

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12 What is the value of $\int_5^5 (x^4 + 3x + 1)\,dx$?

a. 0

b. 1

c. 5

d. Undefined

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